

# Intergenerational Housing Misallocation and Fertility

## Compact Analytical Model

Draft note

June 12, 2026

### Abstract

Children take up space, and the space they take up is scarce in a particular way: family-sized units are concentrated in the owner-occupied segment of the housing market, while buying owner housing requires a down payment and debt-service capacity that many young households do not have. This note develops a compact stationary two-period OLG theory of that friction. Each period, young households decide whether to enter a housing market, choose between renting and owning, choose how many children to have, and save for old age; births, together with a renewal inflow, generate the next young cohort. Young owners become next period's old incumbents, who decide how much of their carried housing to keep before exiting. The model delivers three results. First, every constraint on family-sized space—rental size limits, down payments—collapses into a single object, an effective family-housing cap, and a binding cap raises the price of a child through the space the child requires. Second, the stationary competitive equilibrium clears the housing market and still misallocates the existing stock whenever young households are priced out of owner housing by financing constraints while old incumbents are locked in place by capital-gains taxes that sale triggers and death forgives; the same tax, anticipated, raises the price of owning family-sized space for the young. Both gaps are measurable, one from household bundles and one from deeds and tax schedules. Third, at fixed prices, a housing policy changes aggregate fertility through constrained exposure, pass-through to the binding cap, the household fertility response, entry, and tenure switching.

The market structure follows [Coven et al. \(2025\)](#): tenure shapes which size limits and financing constraints a household faces, while all floorspace clears in a single market; supply margins are deferred to an extension. Children require housing space, so anything that rations family-sized space raises the private price of children. Family-sized space is rationed by tenure: by the size composition of the rental stock on one side, and by down-payment requirements on the other. The old-side force—capital-gains taxes that fall due on sale but are forgiven at death—amplifies the scarcity but is not needed for any of the young-side results.

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## Part I

# Compact Analytical Model

## 1 Environment

**Agents.** The economy is populated by overlapping generations of young and old households. A young household has type

$$i = (y_i, a_i),$$

where  $y_i > 0$  is disposable nonhousing income and  $a_i \geq 0$  is wealth at entry, inherited from the old who exit as she arrives. She decides whether to enter the housing market and, inside, chooses tenure, occupied housing, fertility, and savings. Old households, indexed by  $j$ , come in two groups. Old incumbents are last period's young owners: each enters old age with financial wealth, the house she bought when young, and a possible accrued capital gain; she chooses consumption, how much housing to retain, and how large an estate to leave, then exits. Old renters are last period's young renter entrants; they rent in old age, carry no housing and no gain, and exit. Nonentrants live outside the modeled market in both periods.

**Masses.** Time is discrete. At the start of period  $t$  a mass  $M_t > 0$  of potential young households is drawn from a distribution  $G_{Y,t}$  over types. Next period's potential young are the maturing children of the current cohort plus a renewal inflow of mass  $R_t$ ; entry then determines how many of them participate in the housing market. The period- $t$  measures of old incumbents and old renters are  $F_{O,t}$  and  $F_{R,t}$ . Let  $\bar{M}$  denote the stationary cohort mass. In a stationary equilibrium,  $G_{Y,t} = G_Y$ ,  $F_{O,t} = F_O$ , and  $F_{R,t} = F_R$  are reproduced by the transition laws below.

**Preferences.** Households have preferences over consumption, housing, and children. Children raise the household's consumption and housing needs: each child claims  $\chi > 0$  units of the consumption good and  $\kappa > 0$  units of housing services, so with consumption spending  $c_i$ , occupied housing  $h_i$ , and fertility  $n_i$ , young preferences are

$$u_i^Y(c_i, h_i, n_i) = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i, \quad \alpha, \vartheta > 0.$$

One more child means less space and less consumption left for the parents: children compete with their parents for the same house and the same budget. Because  $\vartheta \log n_i$  rules out  $n_i = 0$ , every entered household has positive completed fertility: the model studies the intensive margin, not childlessness.

Old preferences are

$$u^O(c_j^O, h_j^O, e_j) = \log c_j^O + \gamma \log h_j^O + \mathcal{B}(e_j), \quad \gamma > 0,$$

over consumption, retained housing, and the estate  $e_j \geq 0$  the household leaves when she exits, valued through a warm-glow function  $\mathcal{B}$ , increasing and concave, possibly identically zero, and common to all old households. The exiting cohort's estates fund the entry wealth of the cohort that arrives, through the transition below. A young household also values her own old age: her objective adds  $\beta u^O$ , evaluated at the old-age choices she will make, with discount factor  $\beta \in (0, 1]$ ; the old state she will hold is generated by the transition below.

**Housing.** There is a fixed aggregate stock  $\bar{H}$  of divisible floorspace, measured in units of housing services. A household occupies floorspace in one of two tenure modes: renting, denoted  $R$ , or owning,

denoted  $O$ . The modes offer different size limits, the largest sizes available to rent and to own:

$$\mathcal{H}^R = \{h : 0 < h \leq \bar{h}^R\}, \quad \mathcal{H}^O = \{h : 0 < h \leq \bar{h}^O\}, \quad 0 < \bar{h}^R < \bar{h}^O.$$

Family-sized houses are mostly available to buy, not to rent: a household that wants substantially more space than  $\bar{h}^R$  must buy it.<sup>1</sup> All occupied floorspace trades in one market. The asset price of a unit of floorspace is  $P$ , and a no-arbitrage user-cost condition ties the rental rate to the price:

$$q = \rho P, \quad \rho = r + \delta + \tau^p, \quad (1)$$

where  $r$  is the interest rate,  $\delta$  is depreciation, and  $\tau^p$  is a recurring property tax. Renters and owner-occupiers therefore face the same per-unit flow cost  $q$  on new occupancy choices; tenure matters through the size limits and through financing, not through the per-unit price of space, and the capital-gains tax below shifts only old incumbents' retention margin. Given  $q$ , a higher property tax lowers the asset price  $P$  and hence the down payment needed to buy a given amount of space. Throughout, lowercase  $h$  denotes an occupancy choice; capital  $H$  denotes housing quantities not chosen in the current period—the stock  $\bar{H}$ , effective caps, carried housing, aggregate demand. At the start of a stationary period, old incumbents carry into old age the houses generated by previous young owner choices, and passive landlords hold the residual stock:

$$H_{\text{pass}} = \bar{H} - \int H_j^0 dF_O(j) \geq 0.$$

In a stationary equilibrium the inequality is implied by market clearing: the previous cohort's owner purchases cannot exceed the stock. Ownership of the stock is thus split between old incumbents and passive landlords: young buyers purchase from incumbents' released units and the landlord stock, renters occupy landlord space, and rents and sale proceeds are transfers. Housing payments are transfers across households, landlords, and fiscal accounts, closed by lump-sum rebates, and reallocating the existing stock costs no real resources.<sup>2</sup>

**Financing.** Buying floorspace requires financing. A young buyer can finance at most a share  $\phi \in (0, 1)$  of the purchase; the rest, the down payment  $(1 - \phi)Ph$ , is cash at closing, posted from wealth in hand: the mortgage is what lenders advance against the house, and they advance nothing against future income, so  $a_i$  caps what a buyer can put down. Wealth is otherwise fungible. It enters the young budget below alongside income and finances consumption and saving; buying assets is not expenditure, so the posted equity remains part of the household's wealth. Inheritances therefore move both access to owner housing and lifetime resources. Households save: a young household carries  $a'_i \geq 0$  into old age at the exogenous gross return  $1 + r$ , and cannot borrow unsecured against old age. A renter, with no asset to pledge, therefore cannot borrow at all; a buyer's only debt is the financed share  $\phi$  of the purchase. Entry wealth is assigned from the exiting old's estates by the young-type transition below. The interest rate  $r$  is exogenous: households borrow and lend with a passive financial sector, so no domestic capital-market clearing condition is imposed, and the only price determined inside the model is  $q$ .

<sup>1</sup>Coven et al. (2025) model the same segmentation as a minimum size for owned homes, with rentals available in any size; we cap rental size instead. Both make family-sized space owner-concentrated; ours keeps the renter problem interior at the bottom.

<sup>2</sup>Two supply extensions are deliberately left out of the baseline and developed later: a price-elastic aggregate supply curve as in Coven et al. (2025), and tenure-specific stocks with separate construction margins, under which the rental and owner user costs diverge endogenously. The local results below survive with  $q$  replaced by the relevant tenure-specific user cost.

**Capital-gains taxation.** The model carries two taxes: the property tax  $\tau^p$  inside the user cost, and the capital-gains tax, whose treatment of death is the old-side institution at the center of the model. Selling a house triggers a tax: the seller pays the rate  $\tau^g$  on the difference between the sale price and what she paid, in dollars. Dying instead passes the house with a stepped-up basis—the heir’s purchase price is reset to the current value—so the gain is never taxed. For an old incumbent the difference between the two disposal routes is a per-unit tax  $\bar{\ell}$  on housing sold while alive.

Where do gains come from when the real price never moves? From inflation. Dollar prices and incomes all grow at rate  $\varpi > 0$ , so relative prices and real choices are unaffected; but the tax form compares dollars from different dates, and a house bought one period ago shows a paper gain even though its real value is unchanged. The tax on that paper gain is real,

$$\bar{\ell} = \tau^g \frac{P \varpi}{1 + \varpi}$$

per unit sold, and this is the model’s only contact with inflation: budgets, prices, and child requirements are real. Every cohort carries the same gain, so every incumbent’s unit bears  $\bar{\ell}$  if sold while alive.<sup>3</sup> Everything is deterministic: a buyer knows the tax her future self will face, and Section 7 prices that anticipation. An estate tax, by contrast, would fall on the transfer in any form and leave the retention margin undistorted; the gains tax distorts it precisely because death forgives it. Estate-tax instruments live in the quantitative model.<sup>4</sup>

**Demography and transitions.** The period timing is: potential young types are drawn; young households decide whether to enter; entrants choose tenure, housing, fertility, and savings; old incumbents choose consumption, retained housing, and estates; old households exit; births and the renewal inflow form the next potential-young cohort; current young owners become next period’s old incumbents. Let

$$\Gamma_O(dj' \mid i, h, a'; q, \tau^p)$$

be the old-incumbent transition for a current young owner of type  $i$  who buys  $h$  and carries savings  $a'$ . It assigns the house bought when young,  $H_{j'}^0 = h$ ; financial wealth

$$a_{j'} = (1 + r) a' - q H_{j'}^0,$$

so that total old wealth  $a_{j'} + q H_{j'}^0$  is exactly the savings’ return—the nominal gain is not real wealth. Housing wealth is in per-period goods units: the old live one more period, so a carried unit is worth one period of services  $q$  to its holder, and the shell that outlives her recirculates through the stock. The asset price  $P$  appears only in the down-payment requirement. Mortgage balances from youth are netted inside  $a_{j'}$ : a household that carried a mortgage has lower financial wealth and the same total. The draw does not condition on fertility: estates go to the arriving cohort, not to own children, so the model has no old-age-support motive for births. Young renter entrants map into old-renter states through a parallel kernel  $\Gamma_R(dj' \mid i, a')$ , which assigns  $a_{j'} = (1 + r) a'$ ; nonentrants receive zero mass under both kernels. Old-renter states follow the same law as (2) with  $\Gamma_R$  and the renter indicator. Given current young choices, the next old-incumbent measure satisfies

$$F_{O,t+1}(A) = M_t \int \pi_{i,t}^E \mathbf{1}\{o_{i,t} = O\} \Gamma_O(A \mid i, h_{i,t}^O, a_{i,t}^{O'}, q_t, \tau_t^p) dG_{Y,t}(i) \quad (2)$$

<sup>3</sup>The statutory exclusion, which exempts gains below a dollar threshold, is an institutional detail we suppress; with it, small holdings escape the tax and lock-in sorts by size.

<sup>4</sup>Assessment protection of the Proposition 13 type, under which incumbents are taxed on below-market assessed values until they move, generates the same per-unit retention wedge.

for every measurable set  $A$  of old states. In stationarity, this measure equals  $F_O$ .

Let

$$n_{i,t}^Y = \mathbf{1}\{o_{i,t} = R\}n_{i,t}^R + \mathbf{1}\{o_{i,t} = O\}n_{i,t}^O \quad (3)$$

denote fertility in the chosen inside case, and let  $\bar{n}_i^E \geq 0$  denote fertility in the outside option. The baseline formulas set  $\bar{n}_i^E = 0$ , but the demographic law allows outside fertility or inflow to renew the population.

Young type  $i' = (y_{i'}, a_{i'})$  is generated by fertility, old estates, and the renewal inflow. Let

$$\Gamma_Y(\mathrm{d}i' \mid i, m, n, F_O, F_R, e; \Upsilon), \quad m \in \{R, O, E\},$$

be the young-type kernel for a birth associated with current type  $i$ , case  $m$ , fertility  $n$ , old bequest policy  $e$ , and renewal law  $\Upsilon$ . The old who exit at the end of period  $t$  endow the cohort that arrives at  $t+1$ : the kernel assigns their estates as the entrants' wealth  $a_{i'}$  and supplies the income components of young types. In stationarity,  $\Gamma_Y$  is restricted so that assigned entry wealth aggregates to the estates of all exiting old households, incumbents and renters:

$$\bar{M} \int \left[ \pi_i^E n_i^Y \mathbb{E}[a_{i'} \mid i, o_i] + (1 - \pi_i^E) \bar{n}_i^E \mathbb{E}[a_{i'} \mid i, E] \right] \mathrm{d}G_Y(i) = \int e_j \mathrm{d}(F_O + F_R)(j),$$

where the expectations are over  $\Gamma_Y$  and renewal draws from  $\Upsilon$  receive no estates. How estates are assigned across entrants is otherwise unrestricted: equal division, random assignment, and lineage rules are all admissible kernels, and every result below conditions on the realized  $G_Y$ ; the quantitative model disciplines the choice by matching the joint distribution of income and inheritances. Let  $R_t \geq 0$  be the renewal inflow, outside births or immigration, with type law  $\Upsilon$ . The next young cohort satisfies, for every measurable set  $A$  of young types,

$$\begin{aligned} M_{t+1} G_{Y,t+1}(A) = M_t \int & \left[ \pi_{i,t}^E n_{i,t}^Y \Gamma_Y(A \mid i, o_{i,t}, n_{i,t}^Y, F_{O,t}, F_{R,t}, e_t; \Upsilon) \right. \\ & \left. + (1 - \pi_{i,t}^E) \bar{n}_i^E \Gamma_Y(A \mid i, E, \bar{n}_i^E, F_{O,t}, F_{R,t}, e_t; \Upsilon) \right] \mathrm{d}G_{Y,t}(i) + R_t \Upsilon(A). \end{aligned} \quad (4)$$

Setting  $A$  equal to the whole young type space gives the cohort-mass law:

$$M_{t+1} = M_t \int \left[ \pi_{i,t}^E n_{i,t}^Y + (1 - \pi_{i,t}^E) \bar{n}_i^E \right] \mathrm{d}G_{Y,t}(i) + R_t. \quad (5)$$

In a stationary equilibrium,  $M_t = \bar{M}$ ,  $G_{Y,t} = G_Y$ ,  $F_{O,t} = F_O$ ,  $F_{R,t} = F_R$ , and  $R_t = \bar{R}$ . If  $\bar{R} = 0$ , stationarity requires average births per potential young household to be one. If  $\bar{R} > 0$ , it requires average births below one, with the shortfall covered by renewal.

**Entry.** A potential young household can enter the housing market or take an outside option with lifetime value  $\bar{W}^E$ ; entry is subject to Type-I extreme-value taste shocks with scale  $\kappa_E > 0$ .

## 2 The young household's problem

Conditional on entry, household  $i$  takes the user cost  $q$  and the asset price  $P = q/\rho$  as given and chooses a tenure mode  $m \in \{R, O\}$ , a bundle, and savings. For  $m \in \{R, O\}$ ,

$$\begin{aligned} W_i^m(q, \tau^p) = \max_{c_i, h_i, n_i, a_i'} & \left\{ \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i \right. \\ & \left. + \beta [\mathbf{1}\{m = R\} V^R(a_{i'}) + \mathbf{1}\{m = O\} V^O(a_{i'}, h_i)] \right\} \end{aligned} \quad (6)$$

subject to

$$c_i + qh_i + a'_i = y_i + a_i, \quad (7)$$

$$h_i \in \mathcal{H}^m, \quad a'_i \geq 0,$$

$$a_{j'} = (1 + r) a'_i - \mathbf{1}\{m = O\} qh_i, \quad (8)$$

$$\mathbf{1}\{m = O\} (1 - \phi) Ph_i \leq a_i, \quad (9)$$

$$c_i - \chi n_i > 0, \quad h_i - \kappa n_i > 0.$$

$V^O$  and  $V^R$  are the old-age values defined in the next section, evaluated at the state the household generates. Nothing here is random: the household knows the old state her choices produce. The budget is the same in both modes—space costs  $q$  per unit either way, wealth is spendable either way, savings earn  $1 + r$  either way—so tenure matters through the size limit and the two owner financing requirements, not through the market price of space. Children have a market price. Each child claims  $\chi$  goods and  $\kappa$  units of space, and space rents at  $q$ , so providing for a child costs  $\chi + \kappa q$  per period; the budget splits into the adults' own consumption and space, the children's cost, and savings. When access to space is constrained, the space part of a child's cost is valued above  $q$ —Section 7 makes this exact. A useful benchmark is equal scaling,  $\chi = \kappa q / \alpha$ : the child splits her cost between space and goods in the proportion  $\kappa q / \chi = \alpha$ , the same proportion in which her parents split their own spending.

Using  $P = q / \rho$ , owner housing is bounded above by

$$h_i \leq H_i^O(q, \tau^p) = \min \left\{ \bar{h}^O, \frac{a_i \rho}{(1 - \phi)q} \right\}.$$

The renter analog is  $H_i^R = \bar{h}^R$ . In mode  $m$ ,  $H_i^m$  is the effective family-housing cap. A renter is capped by the rental size limit. A would-be buyer is capped by the owner size limit or by down-payment capacity, whichever is shorter. A policy affects fertility through this channel only when it relaxes the binding component of  $H_i^m$ . If the owner case has no feasible allocation, for example when  $a_i = 0$ , set  $W_i^O = -\infty$ .

The inside value of entry is

$$W_i^Y(q, \tau^p) = \max\{W_i^R(q), W_i^O(q, \tau^p)\}. \quad (10)$$

Let  $o_i \in \{R, O\}$  denote an optimal tenure.<sup>5</sup>

### 3 The old household's problem

The baseline young-access mechanism below does not require any old-side force. When the old side is active, the compact model uses the capital-gains institution from the environment. On old-household objects ( $V^O$ ,  $V^R$ ,  $u_j^O$ ,  $c_j^O$ ) the superscripts mark the old stage and its tenure; the index  $j$  separates old households from young ones. Old incumbent  $j$  is last period's young owner and enters with financial wealth  $a_j$  (negative when a mortgage balance remains), carried housing  $H_j^O$ , and an accrued gain on it. Old households have no income: resources are returned savings and,

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<sup>5</sup>The hard rental cap  $\bar{h}^R$  is the simplest representation of size segmentation. A smooth alternative lets renters pay  $R(h) = qh + \varrho(h)$  with  $\varrho_h(h) \geq 0$  for large units, so that large rentals exist but at a premium; every marginal condition below goes through with  $q$  replaced by  $R_h(h)$ . Such a premium is also one natural source of a rent-own gap in per-unit costs, which Section 9 studies.

for incumbents, the carried house. Each unit she holds faces one choice: sell it now and pay the gains tax, or keep it until death, when the basis steps up and the tax is forgiven. Selling triggers the tax  $\bar{\ell}$  per unit.<sup>6</sup> She chooses consumption, retained housing, and her estate, with the preferences of Section 1:

$$\begin{aligned} V^O(a_j, H_j^0) &= \max_{c_j^O, h_j^O, e_j} u^O(c_j^O, h_j^O, e_j) \\ \text{subject to } & c_j^O + e_j + q h_j^O = a_j + q H_j^0 - \bar{\ell}(H_j^0 - h_j^O), \quad 0 \leq h_j^O \leq H_j^0. \end{aligned} \quad (11)$$

Releasing a unit of housing nets  $q - \bar{\ell}$ , its market value less the tax the sale realizes, while the market values the released space at  $q$ . The forgiveness at death is a subsidy to staying put: a tax rule attached to the disposal route, not a preference and not a real resource saving. The planner below counts true old housing utility and true warm-glow bequest utility but does not count the avoided tax as a social benefit. An old household that keeps a large house because she values it is using housing efficiently; an old household that keeps it because selling would realize a taxable gain is not.

An old renter carries no housing and no tax: her wealth is her returned savings,  $a_j = (1 + r)a'_i$ , positive because zero savings would leave her nothing to consume. With the same preferences she solves

$$V^R(a_j) = \max_{c_j^O, h_j^O, e_j} u^O(c_j^O, h_j^O, e_j) \quad \text{subject to} \quad c_j^O + e_j + q h_j^O = a_j, \quad 0 < h_j^O \leq \bar{h}^R, \quad (12)$$

and at an interior choice  $\gamma c_j^O / h_j^O = q$ : old rental demand adds to market clearing but carries no wedge.<sup>7</sup>

## 4 Entry

With the entry technology of Section 1, the probability that type  $i$  enters the housing market is

$$\pi_i^E(q, \tau^p) = \frac{\exp\{W_i^Y(q, \tau^p)/\kappa_E\}}{\exp\{W_i^Y(q, \tau^p)/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}. \quad (13)$$

The endogenous measure of young entrants is

$$dF_Y^E(i; q, \tau^p) = \bar{M} \pi_i^E(q, \tau^p) dG_Y(i).$$

The limit  $\kappa_E \downarrow 0$  gives deterministic entry, with the indifference set  $W_i^Y = \bar{W}^E$  assumed to carry zero mass. Conditional on entry, the household chooses tenure according to (10).

## 5 Competitive equilibrium

A stationary equilibrium clears one aggregate floorspace market and closes the old-incumbent, young-type, and cohort-mass fixed points. Households take the user cost as given, choose tenure

<sup>6</sup>The statutory tax is triggered by the sale of the discrete dwelling; with divisible floorspace the per-unit tax is its smooth counterpart. Discreteness concentrates the same wedge at the downsizing margin and strengthens retention.

<sup>7</sup>Coven et al. (2025) model this lock-in channel in lifecycle form and evaluate a rate reduction. The tax  $\bar{\ell}$  prices the same margin statically, and also prices the opposite reform: taxing accrued gains at every transfer, including bequest, removes the step-up and shrinks  $\bar{\ell}$  to the value of tax deferral.



and housing, and their choices aggregate into demand for the fixed stock. There is no pairwise assignment of old units to young households: one price equates total demand to the stock, and nothing in that condition asks whether the marginal family-sized unit is held by the household that values it most. That question is deferred to the efficiency analysis below.

Given the policy  $\tau^p$ , the stock  $\bar{H}$ , the stationary cohort mass  $\bar{M}$ , young distribution  $G_Y$ , old measures  $F_O$  and  $F_R$ , and transfer rules, aggregate floorspace demand  $H^D(q, \tau^p, x; \bar{M}, G_Y, F_O, F_R)$  at a candidate user cost  $q$  adds young households, old incumbents, and old renters:

$$H^D = \bar{M} \int \pi_i^E [\mathbf{1}\{o_i = R\} h_i^R + \mathbf{1}\{o_i = O\} h_i^O] dG_Y(i) + \int h_j^O dF_O(j) + \int h_j^O dF_R(j).$$

Tenure determines which size limit and which financing constraints apply to a household, not which market it buys space in: all of this demand falls on the same stock.

**Definition 1** (Stationary competitive equilibrium). *Given  $(\tau^p, \bar{H})$ , transition laws  $(\Gamma_O, \Gamma_R, \Gamma_Y)$ , and a stationary renewal law  $(\bar{R}, \Upsilon)$ , a stationary competitive equilibrium consists of a stationary cohort mass  $\bar{M} > 0$ , a user cost  $q$ , an asset price  $P$ , a stationary young distribution  $G_Y$ , a stationary old-incumbent measure  $F_O$ , a stationary old-renter measure  $F_R$ , young case allocations including savings, tenure choices, entry probabilities, old allocations, and fiscal transfers such that:*

1. *conditional on entry, young households solve (6) in each mode, then choose tenure by (10);*
2. *old incumbents solve (11) and old renters solve (12);*
3. *entry is given by (13);*
4. *the asset price satisfies the no-arbitrage condition (1);*
5. *the single floorspace market clears,*

$$H^D(q, \tau^p, x; \bar{M}, G_Y, F_O, F_R) = \bar{H}; \quad (14)$$

6. *passive housing-income, credit, property-tax, capital-gains, and policy-instrument accounts are closed by lump-sum transfers;*
7. *the old-incumbent measure is stationary: for every measurable set  $A$  of old states,*

$$F_O(A) = \bar{M} \int \pi_i^E \mathbf{1}\{o_i = O\} \Gamma_O(A \mid i, h_i^O, a_i^{O'}; q, \tau^p) dG_Y(i), \quad (15)$$

*and the old-renter measure satisfies the same condition with  $\Gamma_R$  and  $\mathbf{1}\{o_i = R\}$ ;*

8. *the young cohort is stationary: for every measurable set  $A$  of young types,*

$$\begin{aligned} \bar{M} G_Y(A) = & \bar{M} \int \left[ \pi_i^E n_i^Y \Gamma_Y(A \mid i, o_i, n_i^Y, F_O, F_R, e; \Upsilon) \right. \\ & \left. + (1 - \pi_i^E) \bar{n}_i^E \Gamma_Y(A \mid i, E, \bar{n}_i^E, F_O, F_R, e; \Upsilon) \right] dG_Y(i) + \bar{R} \Upsilon(A), \end{aligned} \quad (16)$$

*where  $e$  is the old bequest-expenditure policy. Taking  $A$  to be the whole young type space gives the stationary cohort-balance condition*

$$\bar{M} = \bar{M} \int [\pi_i^E n_i^Y + (1 - \pi_i^E) \bar{n}_i^E] dG_Y(i) + \bar{R}. \quad (17)$$

Let  $\Xi^\Pi \geq 0$  denote real resource costs of policy instruments. The transfer rule is chosen so that, after summing household budgets and using (14), the aggregate goods constraint is

$$\begin{aligned} \bar{M} \int \pi_i^E [\mathbf{1}\{o_i = R\}(c_i^R + a_i'^R - y_i - a_i) + \mathbf{1}\{o_i = O\}(c_i^O + a_i'^O - y_i - a_i)] dG_Y(i) \\ + \int (c_j^O + e_j - a_j) d(F_O + F_R)(j) + \Xi^\Pi \leq 0. \end{aligned}$$

Housing quantities are disciplined separately by floorspace feasibility: housing payments are transfers that net out against rebates, and the old carried-housing term  $qH_j^0$  is a claim on the existing stock rather than new output. The intergenerational books close through the transitions: entry wealth nets against old estates by the adding-up restriction on  $\Gamma_Y$ , and the net return  $r$  on carried savings is paid by the passive financial sector, the open-economy counterpart of foreign asset income. Housing in this baseline is a pure allocation problem.

Aggregate births generated by the potential-young cohort are

$$N = \bar{M} \int [\pi_i^E n_i^Y + (1 - \pi_i^E) \bar{n}_i^E] dG_Y(i).$$

In a stationary demographic equilibrium,  $N + \bar{R} = \bar{M}$ . Under the baseline outside-option normalization  $\bar{n}_i^E = 0$ ,  $N = \bar{M} \int \pi_i^E n_i^Y dG_Y(i)$ . Everything here is the stationary state: the old cross-section is the one the model itself regenerates, so no separate initial old generation needs specifying; an initial condition would matter only on a transition path, which we do not study.

## 6 Constrained planner

The planner comparison is evaluated at a stationary competitive equilibrium. For the local variations below, the stationary cohort mass  $\bar{M}$ , young distribution  $G_Y$ , old measures  $F_O$  and  $F_R$ , transition laws  $(\Gamma_O, \Gamma_R, \Gamma_Y)$ , renewal inflow  $\bar{R}$ , and tax rules are held fixed at the instant of the variation. The question is which constraints young families face as real scarcity and which are financing arrangements that a different institutional design could undo.

The planner has the same preferences, stationary young distribution, old measures  $F_O$  and  $F_R$ , entry technology, housing stock, and tenure size limits as the competitive economy. It chooses entry-relevant inside allocations, tenure assignments, young consumption, housing and fertility, old consumption, housing and estates, and policy instruments for the current stationary cross-section. It does not move the savings margin: young savings  $a_i'$ , and the old wealth they generate, are held at their equilibrium values, so the comparison is within-period reallocation given the intergenerational wealth transition. The planner respects floorspace feasibility and the rental and owner size limits.

The planner faces the economy's real constraints and nothing else: the floorspace stock, the tenure size limits, goods feasibility, and the domain conditions below. It does not face household budgets, since allocations are supported by lump-sum transfers, and it does not face the financing inequalities, which restrict market purchases rather than occupancy; when an allocation must be implemented through market purchases, Section 8 prices the required instruments by their marginal real cost. The planner counts true old housing utility and true warm-glow bequest utility; it does not count the avoided gains tax  $\bar{\ell}$ , which is a transfer.

Let  $\omega_i^Y > 0$  and  $\omega_j^O > 0$  be Pareto weights. Let  $\mathcal{G}_i(n_i)$  denote an optional social payoff from births beyond parental utility. The fertility-neutral planner has  $\mathcal{G}_i \equiv 0$ .

If the planner assigns young type  $i$  tenure  $o_i^P \in \{R, O\}$  and allocation  $(c_i, h_i, n_i)$ , the inside value relevant for entry is

$$V_i^P = u_i^Y(c_i, h_i, n_i).$$

The induced entry probability is

$$\pi_i^P = \frac{\exp\{V_i^P/\kappa_E\}}{\exp\{V_i^P/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}.$$

The corresponding expected entry value, up to an additive constant, is

$$\mathcal{W}_i^E(V_i^P) = \kappa_E \log [\exp\{V_i^P/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}].$$

The planner solves

$$\max \bar{M} \int \omega_i^Y \mathcal{W}_i^E(V_i^P) dG_Y(i) + \bar{M} \int \pi_i^P \mathcal{G}_i(n_i) dG_Y(i) + \int \omega_j^O u^O(c_j^O, h_j^O, e_j) d(F_O + F_R)(j) \quad (18)$$

subject to floorspace feasibility,

$$\bar{M} \int \pi_i^P [\mathbf{1}\{o_i^P = R\} + \mathbf{1}\{o_i^P = O\}] h_i dG_Y(i) + \int h_j^O d(F_O + F_R)(j) \leq \bar{H},$$

size-limit feasibility,

$$h_i \in \mathcal{H}^R \text{ if } o_i^P = R, \quad h_i \in \mathcal{H}^O \text{ if } o_i^P = O,$$

and goods feasibility,

$$\begin{aligned} & \bar{M} \int \pi_i^P (c_i + a'_i - y_i - a_i) dG_Y(i) \\ & + \int (c_j^O + e_j - a_j) d(F_O + F_R)(j) + \Xi^\Pi \leq 0, \end{aligned}$$

All allocations also satisfy

$$c_i - \chi n_i > 0, \quad n_i > 0, \quad h_i - \kappa n_i > 0,$$

for young households, and

$$c_j^O > 0, \quad e_j \geq 0, \quad 0 \leq h_j^O \leq H_j^0,$$

for old incumbents, with the retention bound replaced by the rental size limit for old renters.

Let  $\Lambda$  be the multiplier on goods feasibility and  $\Lambda Q$  the multiplier on floorspace feasibility, so that  $Q$  is the planner's scarcity price of space in consumption goods. For a fixed tenure assignment and active set, and away from size-limit corners, an interior planner allocation satisfies

$$\begin{aligned} \frac{u_{h,i}^Y}{u_{c,i}^Y} &= Q && \text{for young households,} \\ \frac{u_{h,j}^O}{u_{c,j}^O} &= Q && \text{for old housing,} \end{aligned}$$

$$\frac{\vartheta(c_i - \chi n_i)}{n_i} + \nu_i^g = \chi + \kappa Q,$$

$$\frac{\mathcal{B}'(e_j)}{u_{c,j}^O} = 1 \quad \text{if } e_j > 0.$$

The planner equalizes every household's housing-consumption tradeoff to the common scarcity price of space, prices children at  $\chi + \kappa Q$ , and respects the bequest motive. The term  $\nu_i^g$  is the goods-equivalent social value of a marginal birth beyond parental utility. If  $\mathcal{G}_i \equiv 0$ , then  $\nu_i^g = 0$ . Endogenous logit entry affects the levels of the first-order conditions through a common factor, but that factor cancels from the housing-consumption and fertility ratios displayed here.

Conditional on the entry measure and tenure assignment, the planner's interior allocation solves in one line. With  $\mathcal{G}_i \equiv 0$  and log warm glow  $\mathcal{B}(e) = b \log e$ ,

$$c_i - \chi n_i = \frac{\omega_i^Y}{\Lambda}, \quad h_i - \kappa n_i = \frac{\alpha \omega_i^Y}{\Lambda Q}, \quad n_i = \frac{\vartheta \omega_i^Y}{\Lambda(\chi + \kappa Q)},$$

$$c_j^O = \frac{\omega_j^O}{\Lambda}, \quad h_j^O = \frac{\gamma \omega_j^O}{\Lambda Q}, \quad e_j = \frac{b \omega_j^O}{\Lambda},$$

with  $(\Lambda, Q)$  pinned by goods and floorspace feasibility. The weights are not innocent:  $\omega$  scales each household's consumption and space together, so any statement about who gets more space under the planner than in equilibrium depends on the weights chosen. What no weight can touch is the ratio: every planner household satisfies

$$\frac{h_i - \kappa n_i}{c_i - \chi n_i} = \frac{\alpha}{Q},$$

common across households under any  $\omega$ , while the equilibrium disperses the same ratio to  $\alpha/(q^m + \zeta_i^m)$  across constrained young households and, for old incumbents,  $\gamma/(q - \bar{\ell})$  against the planner's  $\gamma/Q$ . Dispersed ratios cannot be matched by any choice of weights. The local test of Section 8 is this violation stated constructively: between a high-ratio and a low-ratio household, goods are on the table.

## 7 Equilibrium policies

How tightly a constraint binds can be read off the household's bundle: the consumption, housing, and fertility she chooses. A household that is free to adjust equates her marginal value of housing space to the price of space. A household pinned by a size limit or a financing requirement values space above its price, and the distance between the two is the constraint's bite, measured in consumption goods.

Every condition below is a derivative of one Lagrangian. For mode  $m$ ,

$$\begin{aligned} \mathcal{L}_i^m = & \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \vartheta \log n_i + \beta V_{j'} \\ & + \lambda_i^m (y_i + a_i - c_i - q h_i - a'_i) + \theta_i^m (\bar{h}^m - h_i) + \xi_i a'_i \\ & + \mathbf{1}\{m = O\} \mu_i (a_i - (1 - \phi) P h_i), \end{aligned}$$

with  $\lambda_i^m$  on the budget (7),  $\theta_i^m \geq 0$  on the size limit, and  $\mu_i \geq 0$  on the down-payment constraint (9), active only for buyers. The generated old state (8) is substituted into  $V_{j'}$ , and  $\xi_i \geq 0$  is the

multiplier on the no-borrowing bound  $a'_i \geq 0$ —for a renter, the constraint that she cannot borrow at all; a buyer’s only debt is the mortgage. The domain conditions hold strictly at any optimum.

For a renter ( $m = R$ ), at an interior allocation with respect to consumption and fertility,

$$\begin{aligned}\frac{1}{c_i^R - \chi n_i^R} &= \lambda_i^R, \\ \frac{\alpha}{h_i^R - \kappa n_i^R} &= \lambda_i^R q + \theta_i^R, \\ \frac{\vartheta}{n_i^R} &= \frac{\chi}{c_i^R - \chi n_i^R} + \frac{\alpha \kappa}{h_i^R - \kappa n_i^R}.\end{aligned}\tag{19}$$

Converting the cap multiplier into consumption goods,  $\zeta_i^R = \theta_i^R / \lambda_i^R \geq 0$ , the renter’s marginal value of housing is

$$\frac{u_{h,i}^Y}{u_{c,i}^Y} = \frac{\alpha (c_i^R - \chi n_i^R)}{h_i^R - \kappa n_i^R} = q + \zeta_i^R.\tag{20}$$

A renter against the size cap values one more unit of space above the rent it would command, and the gap is computable from the observed bundle as  $\zeta_i^R = \alpha(c_i^R - \chi n_i^R)/(h_i^R - \kappa n_i^R) - q$ , the household’s tradeoff between space and consumption, net of children, in excess of the market rent. An unconstrained renter has  $\zeta_i^R = 0$ .

**Assumption 1** (Regularity at the old margin). *The household’s induced old retention choice is interior,  $h_{j'}^O < H_{j'}^0$ .*

Anticipation is a price here, not a friction to assume away. Buying one more unit raises anticipated old wealth by its housing value and costs that value today; it also commits the household to the tax  $\bar{\ell}$  on the marginal unit her old self sells. By the envelope condition  $\partial V^O / \partial H^0 = (q - \bar{\ell}) / c^O$ , which the assumption keeps smooth, the continuation contributes  $-\beta \bar{\ell} / c_{j'}^O$  to the buyer’s housing condition, and the saving condition converts it into goods:

$$\frac{\beta \bar{\ell} (c_i - \chi n_i)}{c_{j'}^O} = \frac{\bar{\ell}}{1 + r}.$$

A buyer therefore pays an effective price

$$q^O \equiv q + \frac{\bar{\ell}}{1 + r},$$

per unit of space—the market price plus her own discounted lock-in tax—while a renter pays  $q$ . The tax acts twice: ex post it locks in the current old, and ex ante it taxes ownership of family-sized space.

For a young owner ( $m = O$ ), at an interior allocation with respect to consumption and fertility,

$$\begin{aligned}\frac{1}{c_i^O - \chi n_i^O} &= \lambda_i^O, \\ \frac{\alpha}{h_i^O - \kappa n_i^O} &= \lambda_i^O q + \frac{\beta \bar{\ell}}{c_{j'}^O} + \mu_i(1 - \phi)P + \theta_i^O, \\ \frac{\vartheta}{n_i^O} &= \frac{\chi}{c_i^O - \chi n_i^O} + \frac{\alpha \kappa}{h_i^O - \kappa n_i^O}.\end{aligned}\tag{21}$$

The same conversion for the owner collects the access constraints, each priced by the primitive that generates it:

$$\zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i^O}(1-\phi)P}_{\zeta_i^{O,F}, \text{ down-payment component}} + \underbrace{\frac{\theta_i^O}{\lambda_i^O}}_{\text{owner-size-limit component}} \geq 0.$$

The first term is the collateral margin: it is positive when the household's wealth  $a_i$  cannot cover the down payment  $(1-\phi)P$  on a larger unit. The second is the owner size size limit. A payment-to-income test would add a parallel financing component to  $\zeta_i^{O,F}$ ; the quantitative model carries one. If the owner size cap does not bind, then  $\theta_i^O = 0$  and  $\zeta_i^O = \zeta_i^{O,F}$ : over and above her effective price, the household is constrained by finance, not by the stock. The young owner's marginal value of owner housing is

$$\frac{u_{h,i}^Y}{u_{c,i}^Y} = \frac{\alpha(c_i^O - \chi n_i^O)}{h_i^O - \kappa n_i^O} = q^O + \zeta_i^O. \quad (22)$$

Both tenure modes share the saving margin

$$\frac{1}{c_i - \chi n_i} = \beta(1+r) \frac{1}{c_{j'}^O} + \xi_i, \quad \xi_i \geq 0, \quad \xi_i a_i' = 0, \quad (23)$$

the first-order condition for saving, with  $c_{j'}^O$  old consumption at the generated state: both old problems have envelope  $\partial V/\partial a_j = 1/c_j^O$ , carried savings pay  $1+r$ , and  $\xi_i > 0$  exactly when the household would like to borrow against old age and cannot. In the benchmark it pins savings in closed form. With logarithmic warm glow,  $\mathcal{B}(e) = b \log e$  with  $b \geq 0$ , old consumption at interior choices is proportional to liquidation wealth, and the condition collapses to the savings rule

$$a_i' = \beta(1+\gamma+b)(c_i - \chi n_i) + \mathbf{1}\{m=O\} \frac{\bar{\ell} h_i}{1+r}, \quad (24)$$

with  $\gamma$  the old housing weight and  $b$  the warm-glow weight; renters drop the second term. In the first term the gross return drops out, the usual log cancellation; the second is the discounted future tax bill, which the household saves in advance: anticipating the lock-in tax is itself a savings motive. The savings load  $k \equiv \beta(1+\gamma+b)$  is old age's claim on each unit of adult consumption; the comparative statics of Section 9 use it as shorthand.

For an old owner at an interior choice of current housing to keep,

$$\frac{u_{h,j}^O}{u_{c,j}^O} = \frac{\gamma c_j^O}{h_j^O} = q - \bar{\ell}. \quad (25)$$

The old owner's valuation sits *below* the market price by exactly the tax: she keeps marginal space she values at less than the market would pay, because selling it would realize the taxable gain. If bequests are interior, then

$$\mathcal{B}'(e_j) = \frac{1}{c_j^O},$$

which matches the planner's bequest condition: leaving wealth to children is a preference the model respects, not a distortion. An old renter at an interior choice has  $u_{h,j}^O/u_{c,j}^O = q$  exactly; her demand is wedge-free.

**Proposition 1** (Constrained family-housing access and fertility). *Fix an entered young household and its optimal tenure case. If the household rents, then*

$$\frac{\vartheta (c_i^R - \chi n_i^R)}{n_i^R} = \chi + \kappa(q + \zeta_i^R).$$

*If the household owns, then*

$$\frac{\vartheta (c_i^O - \chi n_i^O)}{n_i^O} = \chi + \kappa(q^O + \zeta_i^O),$$

*with  $q^O = q + \bar{\ell}/(1+r)$  the buyer's effective price. Thus rental size limits and owner access constraints enter fertility through the housing services required per child.*

*Proof.* For renters, multiply (19) by  $1/\lambda_i^R = c_i^R - \chi n_i^R$  and use (20). For owners, multiply (21) by  $1/\lambda_i^O = c_i^O - \chi n_i^O$  and use (22).  $\square$

A child's space requirement is priced at the household's *own* shadow value of space, not at the market price. In an unconstrained case, the full price of a child is  $\chi + \kappa q$ . For a constrained household it is  $\chi + \kappa(q + \zeta_i^m)$ : children become more expensive exactly where family-sized space is hard to reach, even with market prices unchanged. Two limits anchor the framework: as  $H_i^m \rightarrow \infty$  every household pays the unconstrained child price  $\chi + \kappa q$ , and with  $\kappa = 0$  children claim no space and the price reduces to the standard child-goods cost  $\chi$ . On the financing side, when the collateral component binds,  $\partial H_i^O / \partial \tau^p = a_i / ((1 - \phi)q) > 0$ : a higher property tax lowers the asset price at a given user cost and relaxes the down payment, the capitalization channel of Coven et al. (2025).

If no access or size limit constraint binds in the relevant tenure case, the corresponding access value is zero. For example, an unconstrained young owner satisfies

$$\begin{aligned} c_i^u - \chi n_i^u &= \frac{y_i + a_i}{1 + \alpha + \vartheta + k}, & a_i^u &= k (c_i^u - \chi n_i^u) + \mathbf{1}\{m = O\} \frac{\bar{\ell} h_i^u}{1 + r}, \\ n_i^u &= \frac{\vartheta (y_i + a_i)}{(1 + \alpha + \vartheta + k)(\chi + \kappa q^O)}, \\ h_i^u &= \frac{\alpha (y_i + a_i)}{(1 + \alpha + \vartheta + k) q^O} + \kappa \frac{\vartheta (y_i + a_i)}{(1 + \alpha + \vartheta + k)(\chi + \kappa q^O)}. \end{aligned}$$

The renter analog has the same closed form and is feasible only if  $h_i^u \leq \bar{h}^R$ .

## 8 Efficiency

At a stationary competitive equilibrium, the price makes aggregate demand equal the fixed stock. Market clearing alone is not enough for efficiency. Conditional on the stationary cross-section and transition margins, efficiency also requires that no feasible reassignment would move marginal space toward households who value it more, net of implementation costs. The test below is compensated: reallocate a unit, hold every household at its prior utility with lump-sum transfers, and ask whether goods are left over. A positive residual is a Pareto improvement under any welfare weights—it restates Section 6's ratio violation constructively, harvesting the dispersion that no weight choice can reconcile.

**Proposition 2** (Planner-equilibrium comparison, local). *Consider an interior stationary competitive equilibrium and a planner with the same preferences, housing stock, tenure size limits, stationary young distribution, old measures, and transition laws held fixed at the instant of the variation. Suppose policy accounts are resource-neutral and the planner has no social birth payoff, so  $v_i^g = 0$ . Conditional on the equilibrium entry measure and tenure assignments, and away from real size-limit corners, the competitive allocation solves the planner's conditional allocation problem only if no feasible marginal reallocation of existing floorspace has positive goods-equivalent surplus.*

Let  $r_i^F \geq 0$  denote the marginal real resource cost, in consumption-good units, of relaxing young owner  $i$ 's financial implementation constraints enough to permit one more unit of owner floorspace. If the planner can directly reassign existing floorspace and relieve financing without using real resources, then  $r_i^F = 0$ ; if it must operate through costly financing instruments,  $r_i^F$  prices those real costs; and if the constraint stands for an enforcement primitive no instrument relaxes, the variation is infeasible, the limit  $r_i^F = \infty$ . For a young owner whose owner size limit is slack and whose financial constraints bind, and for an old incumbent at an interior retained-housing choice, the planner's marginal surplus from increasing the young owner's space by one unit and reducing the old incumbent's retained space by one unit is

$$\left( q^O + \zeta_i^{O,F} - r_i^F \right) - (q - \bar{\ell}) = \zeta_i^{O,F} + \frac{\bar{\ell}}{1+r} + \bar{\ell} - r_i^F.$$

The buyer's anticipated tax is a transfer, like the incumbent's realized one, so it belongs in the recoverable surplus. Thus, in the zero-real-cost finance case, any such pair with  $\zeta_i^{O,F} + \bar{\ell}/(1+r) + \bar{\ell} > 0$  prevents the competitive allocation from solving the planner's conditional allocation problem. More generally, a positive young financial access gap is inefficient whenever it can be relaxed at marginal real cost below the value gap relative to the marginal source of space, and a positive retention tax  $\bar{\ell}$  marks an inefficiency whenever the released space has a feasible recipient whose marginal value exceeds the old incumbent's true marginal value.

Conversely, if the financial implementation gaps are zero for all relevant young owners and  $\bar{\ell} = 0$  for old incumbents at interior retention choices, then, conditional on entry, tenure assignments, savings, and real size limit constraints, the competitive allocation satisfies the planner's conditional housing and fertility first-order conditions for suitable Pareto weights.

*Proof.* The result is a direct marginal-variation comparison that preserves aggregate floorspace feasibility. Increase household  $a$ 's occupied floorspace by  $dh$  and reduce household  $b$ 's occupied floorspace by the same  $dh$ , holding entry, tenure assignments, active real size limits, fertility, savings, and stationary transition margins fixed at the instant of the variation. The goods-equivalent first-order welfare effect is

$$\left( \frac{u_{h,a}}{u_{c,a}} - \text{marginal real implementation cost for } a \right) - \frac{u_{h,b}}{u_{c,b}}.$$

For a young owner with a slack owner size limit and binding financial implementation constraints,

$$\frac{u_{h,i}^Y}{u_{c,i}^Y} = q^O + \zeta_i^{O,F}.$$

If relaxing those constraints has marginal real resource cost  $r_i^F$ , the net marginal value of assigning one more unit of space to that owner is  $q^O + \zeta_i^{O,F} - r_i^F$ , and the anticipated-tax component  $\bar{\ell}/(1+r)$



of  $q^O$  is a transfer the planner recovers. For an old incumbent at an interior retention choice,

$$\frac{u_{h,j}^O}{u_{c,j}^O} = q - \bar{\ell}.$$

Subtracting gives

$$\zeta_i^{O,F} + \frac{\bar{\ell}}{1+r} + \bar{\ell} - r_i^F.$$

A positive value is a feasible first-order improvement: transfer goods to hold each household at its pre-variation utility, and the residual  $\left[\zeta_i^{O,F} + \bar{\ell} - r_i^F\right] dh$  is available as surplus. Therefore the competitive allocation cannot solve the planner's conditional allocation problem.

If all implementation-relevant financial gaps and retention taxes are zero, the competitive marginal values of all households away from real size-limit corners coincide with the common market user cost  $q$ . Real size-limit corners are respected by both the competitive economy and the constrained planner. With  $\nu_i^g = 0$ , the competitive fertility condition then matches the planner's fertility condition evaluated at the same scarcity price, and Pareto weights can be chosen to support the competitive consumption allocation.  $\square$

The proposition does not characterize the efficiency of the entry, tenure-assignment, or intergenerational transition margins themselves; those margins are held fixed. It also does not say that every cap is a market failure: real size limits remain real constraints, while financial gaps matter only to the extent that an implementable instrument can reach them at a low enough real resource cost.

The misallocation has two sides. A young household that cannot post a larger down payment values the marginal unit of space at  $q^O + \zeta_i^{O,F}$ , above the market value of that space; the market clears, yet she cannot buy more of it. An old incumbent sitting on a taxable gain keeps the marginal unit while valuing it at  $q - \bar{\ell}$ , below the market value of that space. Moving one unit from the second household to the first would raise welfare by  $\zeta_i^{O,F} + \bar{\ell}/(1+r) + \bar{\ell} - r_i^F$ , measured in consumption goods; in the pure-finance case the  $r_i^F$  term drops. Both pieces are computable from data given the preference parameters: the young gap from bundles, via  $\zeta_i^{O,F} = \alpha(c_i^O - \chi n_i^O)/(h_i^O - \kappa n_i^O) - q - \bar{\ell}/(1+r)$  when the size cap is slack—the bundle gap over market rent net of the known tax term—and the old gap from deeds and the tax schedule, via the purchase basis, the current value, and the rate. The old-side force amplifies the misallocation but is not needed for it: the young-access gap is positive whenever a purchase constraint binds with positive shadow value, with or without the gains tax.

Two qualifications matter. First, old occupancy is not inefficient by itself. An old household that keeps a large house because she values it, or because she intends to leave it, is using housing efficiently; those motives are inside  $u^O$  and the planner counts them. The inefficiency is only the part of retention supported by the forgiveness at death. Second, not every binding constraint is an inefficiency. The size limits are real features of the housing stock: when a renter is pinned at  $\bar{h}^R$ , the planner—facing the same stock—is pinned with her, and the multiplier on the size limit is the shadow value of a real constraint, not a market failure. Changing the size limits requires construction, conversion, or reassignment instruments, whose resource costs belong in  $\Xi^{\Pi}$ . Down-payment requirements are financing arrangements; their efficiency cost depends on whether feasible instruments can reach them.

**A worked example.** A two-type stationary economy puts numbers on every object above. Normalize  $q = 1$  and set  $\rho = 0.25$ , so  $P = 4$ . The young household has  $y = 1$ ,  $a = 0.20$ ,  $\alpha = 0.5$ ,  $\vartheta = 0.4$ ,  $\kappa = 0.3$ , and the equal-scaling child cost  $\chi = \kappa q/\alpha = 0.6$ ; financing is  $\phi = 0.80$ ; the size

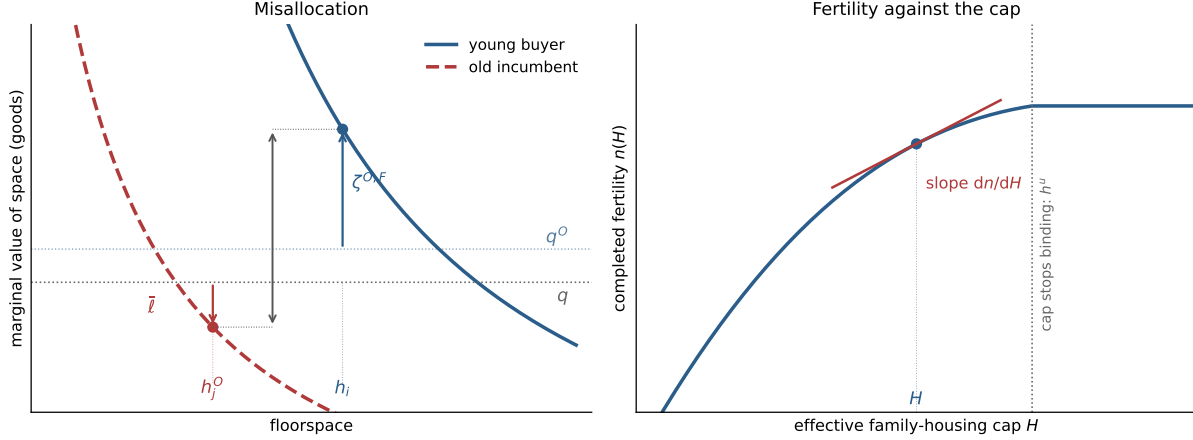


Figure 1: The worked example. Left: marginal value of space. The collateral-capped young owner sits at  $h = 0.25$  valuing space at  $q^O + \zeta^{O,F} = 1.68$ ; the incumbent retains  $h^O = 1/6$  valuing it at  $q - \bar{\ell} = 0.80$ ; moving one unit from her to the buyer frees 0.88 in goods. Right: completed fertility against the effective cap. At the equilibrium cap the slope is  $dn/dH = 0.30$ , and the cap stops binding at  $h^u = 0.31$ .

limits are  $\bar{h}^R = 0.15$  and  $\bar{h}^O = 1$ . The gains tax is  $\bar{\ell} = 0.2$  per unit; the period return is  $r = 0.35$ , so the anticipated-tax wedge is  $\bar{\ell}/(1+r) = 0.148$  and the buyer's effective price is  $q^O = 1.148$ ; the discount factor is  $\beta = 0.1763$ , so with  $\gamma = 0.5$  and no bequest motive the savings load is  $k = \beta(1+\gamma) = 0.264$ . The old incumbent has  $a_j = 0.2$  and carried housing  $H^0 = 0.25$ . The stock  $\bar{H} = 5/12$  is chosen so that  $q = 1$  is the market-clearing user cost; entry is deterministic with the inside value above the outside option, and the renewal inflow  $\bar{R} = 1 - n^*$  sustains the unit cohort.

The young household buys. Both tenures price space at  $q$ , so the case with the larger cap wins: the owner cap is  $H^O = \min\{1, a\rho/((1-\phi)q)\} = \min\{1, 0.25\} = 0.25$ , the collateral component, against the rental size limit 0.15. The cap binds—the unconstrained optimum is  $h^u = 0.312$ —so  $h = 0.25$ . The budget is  $c + qh + a' = y + a = 1.20$ ; the savings rule gives  $a' = k(c - \chi n) + \bar{\ell}h/(1+r) = 0.163 + 0.037 = 0.20$ , so the buyer carries exactly her initial wealth, and the fertility condition gives  $n^* = 0.223$ , with adult consumption  $c - \chi n = 0.616$  and adult space  $h - \kappa n = 0.183$ . The bundle reveals a gap of  $\alpha(c - \chi n)/(h - \kappa n) - q = 0.68$  over market rent: 0.148 of it is her own discounted future tax, and the rest,  $\zeta^{O,F} = 0.53$ , is financing—she would pay above her effective price for one more unit of space and cannot post the down payment to get it. The incumbent's released units net  $q - \bar{\ell}$ , so her liquidation wealth is  $a_j + (q - \bar{\ell})H^0 = 0.40$  and she keeps  $h^O = 1/6$  at the interior solution of  $\gamma c_j^O/h_j^O = q - \bar{\ell}$ , valuing retained space at 0.80; her released 1/12 plus the passive stock 1/6 house the young buyer, and the market clears at  $\bar{H} = 5/12$ . The next old cohort carries the young owner's purchase, 0.25, and her savings return  $(1+r)a'$ , so the stock recirculates; the incumbent's wealth and tax stand in for her cohort's realized repricing, which the buyer does not anticipate. Figure 1 draws the two sides.

Every object can be read off this one allocation: moving one unit of space from the incumbent to the young household frees  $\zeta^{O,F} + \bar{\ell}/(1+r) + \ell = 0.88$  in goods; the cap derivative is  $dn/dH = 0.30$ ; and a down-payment grant passes through at  $\partial H^O/\partial A_i = \rho/((1-\phi)q) = 1.25$ , so a wealth transfer of 0.01 raises the cap by 0.0125 and completed fertility by 0.0038 through access, plus 0.0012 through the budget now that wealth is spendable, 0.0050 in all. The gap, the tax, and the responses are read off the bundle, the tax schedule, and the financing terms.

## 9 Fertility and entry

The fertility implication does not require a planner who directly values births. It follows from the private household problem. Children require consumption and housing services, so a constraint on family-capable housing can change fertility even when fertility enters only parental utility.

Fix a tenure case and suppress the type index. Let  $H$  be the effective upper bound on occupied housing in that case; the user cost is  $q$ . For renters,  $H = \bar{h}^R$ . For young owners,

$$H = \min \left\{ \bar{h}^O, \frac{a_i \rho}{(1 - \phi)q} \right\}.$$

Holding  $(w, \chi, q^m, \alpha, \vartheta, \kappa, k)$  fixed, where  $w \equiv y + a$  is lifetime resources and  $q^m$  is the case's effective price of space— $q$  for renters,  $q + \bar{\ell}/(1 + r)$  for buyers—the case problem adds savings to the bundle choice. Concentrating the savings rule into the objective and the budget reduces it to

$$\max_{c, h, n} \{ (1 + k) \log(c - \chi n) + \alpha \log(h - \kappa n) + \vartheta \log n \}$$

up to a type constant, subject to

$$(1 + k)(c - \chi n) + \chi n + q^m h = w, \quad h \leq H, \quad c - \chi n > 0, \quad h - \kappa n > 0:$$

the static problem with consumption weight  $1 + k$ , resources  $w$ , and the case price  $q^m$ . Each unit of adult consumption commits  $1 + k$  units of resources, one spent now and  $k$  carried for old age; each unit of owner space commits its market cost now plus the discounted tax, saved in advance. The unconstrained case allocation is

$$c^u - \chi n^u = \frac{w}{1 + \alpha + \vartheta + k}, \quad n^u = \frac{\vartheta w}{(1 + \alpha + \vartheta + k)(\chi + \kappa q^m)},$$

$$h^u = \frac{\alpha w}{(1 + \alpha + \vartheta + k) q^m} + \kappa \frac{\vartheta w}{(1 + \alpha + \vartheta + k)(\chi + \kappa q^m)}.$$

Suppose  $0 < H < h^u$ , so the housing cap binds. Then  $h(H) = H$ , and the constrained fertility choice  $n(H)$  is the unique solution to

$$\frac{\vartheta}{n} = \frac{(1 + k) \chi}{w - q^m H - \chi n} + \frac{\alpha \kappa}{H - \kappa n}.$$

Implicit differentiation gives

$$\frac{dn}{dH} = \frac{\frac{\alpha \kappa}{(H - \kappa n)^2} - \frac{(1 + k) \chi q^m}{(w - q^m H - \chi n)^2}}{\frac{\vartheta}{n^2} + \frac{(1 + k) \chi^2}{(w - q^m H - \chi n)^2} + \frac{\alpha \kappa^2}{(H - \kappa n)^2}}.$$

Therefore a relaxation of the binding housing cap raises fertility at  $H$  if and only if

$$\alpha \kappa (w - q^m H - \chi n)^2 > (1 + k) \chi q^m (H - \kappa n)^2.$$

This condition has a direct Stone–Geary interpretation. At an unconstrained interior case allocation,

$$\frac{q^m (h^u - \kappa n^u)}{c^u - \chi n^u} = \alpha,$$

unchanged by the savings margin and evaluated at the case price. The child input bundle has the same housing-to-consumption expenditure ratio as the adults' own bundle when

$$\frac{q\kappa}{\chi} = \alpha, \quad \text{equivalently} \quad \chi = \frac{\kappa q}{\alpha},$$

the equal-scaling benchmark. With savings, the threshold for the fertility response sits above it: fertility is strictly increasing in every binding cap relaxation if and only if

$$\chi \leq (1+k) \frac{\kappa q^m}{\alpha},$$

because forced spending on housing now crowds out saving alongside consumption, and only the share  $1/(1+k)$  of the budget hit lands on the adult-consumption margin that prices children. A binding cap implies  $u_h/u_c = \alpha(c - \chi n)/(H - \kappa n) > q^m$ , so the condition above holds strictly under equal scaling whenever the cap binds. Hence, under equal scaling, every relaxation of a strictly binding family-housing cap raises fertility until the cap ceases to bind. If  $\chi > (1+k)\kappa q^m/\alpha$ , fertility need not rise from every cap relaxation; in particular,  $dn/dH < 0$  for binding caps sufficiently close to the unconstrained optimum. The reason is that a larger cap raises housing slack, which lowers the crowding cost of children, but it also induces the household to buy more housing at price  $q$ , reducing consumption. Fertility rises when the slack effect dominates the consumption-cost effect.

This case result maps into the model's access margins. A relaxation of the rental size cap raises  $H = \bar{h}^R$ . If the owner down-payment constraint is the unique binding component of the effective owner cap, then

$$\frac{\partial H}{\partial A_i} = \frac{\rho}{(1-\phi)q} > 0.$$

The derivative above signs the fertility response to these primitive cap relaxations locally, holding tenure, prices, income, and child costs fixed.

The competitive-equilibrium/planner comparison is a different statement because prices, consumption, tenure assignments, and entry can all change. In the competitive equilibrium, an entered young household in case  $m$  satisfies

$$\frac{\vartheta(c_i^{CE,m} - \chi n_i^{CE,m})}{n_i^{CE,m}} = \chi + \kappa(q^{CE} + \zeta_i^{CE,m}).$$

A fertility-neutral planner has no direct social birth payoff. If the planner allocation for the same type and case is interior with respect to any remaining access constraint, then

$$\frac{\vartheta(c_i^{P,m} - \chi n_i^{P,m})}{n_i^{P,m}} = \chi + \kappa Q^P.$$

Hence

$$n_i^{P,m} > n_i^{CE,m} \iff \frac{c_i^{P,m} - \chi n_i^{P,m}}{c_i^{CE,m} - \chi n_i^{CE,m}} > \frac{\chi + \kappa Q^P}{\chi + \kappa(q^{CE} + \zeta_i^{CE,m})}.$$

Thus a fertility-neutral planner does not mechanically deliver higher fertility merely because the competitive allocation has a positive access value. The case-level result gives a primitive sufficient condition for a signed fertility response when a policy or planner locally raises a type's effective housing cap holding the case primitives fixed.

If the planner directly values births through  $\mathcal{G}_i(n_i)$ , the interior planner condition becomes

$$\frac{\vartheta(c_i^P - \chi n_i^P)}{n_i^P} + \nu_i^g = \chi + \kappa Q^P.$$

A positive  $\nu_i^g$  is a separate fertility motive. Holding consumption, the scarcity price of space, and any remaining access constraints fixed, it raises desired fertility. In the full equilibrium, aggregate fertility also depends on prices, consumption, tenure assignments, and entry.

Under the baseline normalization  $\bar{n}_i^E = 0$ , births generated by the potential-young cohort are

$$N = \bar{M} \int \pi_i^E n_i^Y dG_Y(i).$$

If the outside option has fertility  $\bar{n}_i^E$ , then cohort births are instead

$$N = \bar{M} \int [\pi_i^E n_i^Y + (1 - \pi_i^E) \bar{n}_i^E] dG_Y(i),$$

and every entry-margin term below replaces  $n_i^Y$  by  $n_i^Y - \bar{n}_i^E$ . In the stationary demographic equilibrium, this birth mass and the renewal inflow satisfy  $N + \bar{R} = \bar{M}$ ; the local comparisons below hold  $(\bar{M}, G_Y, F_O, F_R, \bar{R})$  fixed and do not by themselves trace the new stationary demographic fixed point after a policy change. Across two allocations,

$$N^P - N^{CE} = \bar{M} \int [n_i^P(\pi_i^P - \pi_i^E) + \pi_i^E(n_i^P - n_i^{CE})] dG_Y(i),$$

where  $n_i^{CE}$  is fertility in the competitive-equilibrium inside case and outside-option fertility is normalized to zero. The second term is the intensive fertility margin among the households who would enter in the competitive equilibrium. The first term is the entry margin induced by the change in inside value. If entry is held fixed, the aggregate fertility effect is just

$$N^P - N^{CE} = \bar{M} \int \pi_i^E (n_i^P - n_i^{CE}) dG_Y(i).$$

With entry held fixed, aggregate fertility rises only where the planner raises conditional fertility for constrained young types. With entry endogenous, higher inside values can also add entrants.

## 9.1 A local policy decomposition

Let  $z$  be a policy parameter. The local fixed-price effect depends on constrained exposure, pass-through to the binding cap, and the fertility response. For renters,  $H_i^R = \bar{h}^R$ . For owners,

$$H_i^O = \min \left\{ \bar{h}^O, \frac{a_i \rho}{(1 - \phi)q} \right\}.$$

Let  $m \in \{R, O\}$  denote the active tenure case and let  $\mathcal{E}_m(z)$  be the set of types for whom case  $m$  is uniquely active and  $H_i^m(z)$  is the unique binding family-housing cap. Define the policy pass-through to the effective cap by

$$\mathcal{P}_i^m(z) = \frac{\partial H_i^m}{\partial z}.$$

Because the cap is a minimum, a policy that improves a slack component does nothing: a subsidy to units below the binding size does not relax a binding down-payment constraint, and a down-payment

grant does nothing for a renter pinned by the rental size limit. If a policy does not relax the binding component of  $H_i^m$ , then  $\mathcal{P}_i^m(z) = 0$  for this family-space-cap mechanism. Policies that operate primarily through the price  $q$  are a different exercise; they are outside this fixed-price formula, not evaluated by it. A transfer of spendable resources also moves fertility through the budget at a fixed cap; the decomposition isolates the access channel, and an instrument that both relaxes the cap and transfers resources adds the standard income term.

For capped types, the fixed-case fertility response is

$$\Psi_i^m = \frac{\frac{\alpha\kappa}{(h_i^m - \kappa n_i^m)^2} - \frac{\chi q^m}{(1+k)(c_i^m - \chi n_i^m)^2}}{\frac{\vartheta}{(n_i^m)^2} + \frac{\chi^2}{(1+k)(c_i^m - \chi n_i^m)^2} + \frac{\alpha\kappa^2}{(h_i^m - \kappa n_i^m)^2}}.$$

If  $\chi \leq (1+k)\kappa q^m/\alpha$  and the cap binds strictly, then  $\Psi_i^m > 0$ . The boundary case  $\chi = (1+k)\kappa q^m/\alpha$  is equal scaling at lifetime prices: the child's consumption requirement matches its space requirement when adult consumption is priced at its lifetime cost  $1+k$  and space at the case price.

Let

$$\Theta_i^m \equiv \frac{\partial W_i^m}{\partial H_i^m} = \lambda_i^m \zeta_i^m$$

be the utility value of relaxing the effective cap. Holding prices, tenure choices, the active sets  $\mathcal{A}$ —the collection of case assignments and binding cap components—and the stationary demographic state fixed, the local aggregate fertility effect is

$$\left. \frac{dN}{dz} \right|_{q,o,\mathcal{A}} = \bar{M} \sum_{m \in \{R,O\}} \int_{\mathcal{E}_m(z)} \left[ \pi_i^E \Psi_i^m + (n_i^m - \bar{n}_i^E) \frac{\pi_i^E (1 - \pi_i^E)}{\kappa_E} \Theta_i^m \right] \mathcal{P}_i^m(z) dG_Y(i),$$

where  $\bar{n}_i^E = 0$  under the baseline outside-option normalization. The first term is the intensive margin: already-entered constrained households adjust fertility by  $\Psi_i^m$  per unit of cap relief, positive under  $\chi \leq (1+k)\kappa q^m/\alpha$ . The second is the entry margin: relaxing a binding cap raises the value of entering the market, and entrants bring their fertility with them—net of the births they would have had outside. Both margins are gated by the same exposure set  $\mathcal{E}_m(z)$ : a policy aimed at unconstrained households moves nothing.

At the lifetime equal-scaling point, one observable object drives both margins.

**Corollary 1** (Equal scaling and access gaps). *Suppose  $\chi = (1+k)\kappa q^m/\alpha$ , equal scaling at lifetime prices in case  $m$ , and the effective cap binds strictly. Then*

$$\Psi_i^m = \frac{\kappa}{\alpha} \frac{\zeta_i^m}{c_i^m - \chi n_i^m} \Omega_i^m, \quad \Omega_i^m = \frac{\frac{\alpha}{h_i^m - \kappa n_i^m} + \frac{q^m}{c_i^m - \chi n_i^m}}{\frac{\vartheta}{(n_i^m)^2} + \frac{\chi^2}{(1+k)(c_i^m - \chi n_i^m)^2} + \frac{\alpha\kappa^2}{(h_i^m - \kappa n_i^m)^2}} > 0.$$

*Under lifetime equal scaling, the access gap  $\zeta_i^m$  measures both misallocation and fertility exposure. The entry margin runs on the same gap, since  $\Theta_i^m = \zeta_i^m/(c_i^m - \chi n_i^m)$ . Policies aimed at the largest normalized gaps  $\zeta_i^m/(c_i^m - \chi n_i^m)$  raise births the most, holding pass-through, entry, and the weight  $\Omega_i^m$  fixed.*

*Proof.* At  $\chi = (1+k)\kappa q^m/\alpha$ , the numerator of  $\Psi_i^m$  is

$$\frac{\alpha\kappa}{(h_i^m - \kappa n_i^m)^2} - \frac{\kappa(q^m)^2}{\alpha(c_i^m - \chi n_i^m)^2} = \frac{\kappa}{\alpha} \left( \frac{\alpha}{h_i^m - \kappa n_i^m} - \frac{q^m}{c_i^m - \chi n_i^m} \right) \left( \frac{\alpha}{h_i^m - \kappa n_i^m} + \frac{q^m}{c_i^m - \chi n_i^m} \right) :$$

the  $1 + k$  in the child cost cancels the  $1 + k$  in the consumption term of the numerator. At a binding cap,  $\alpha(c_i^m - \chi n_i^m)/(h_i^m - \kappa n_i^m) = q^m + \zeta_i^m$ , so the first parenthesis equals  $\zeta_i^m/(c_i^m - \chi n_i^m)$ . Dividing by the positive denominator gives the expression above.  $\square$

Thus the fixed-case effect is exposure to a binding family-space constraint, times pass-through to the effective cap, times the fertility response, plus the induced entry response. The decomposition also holds tenure assignments fixed. With deterministic tenure choice, a policy that changes  $W_i^O - W_i^R$  moves the indifference boundary  $W_i^O = W_i^R$ . If the type distribution has density near that boundary and fertility differs across cases, this tenure-switching margin is first order. The formula above is therefore the fixed-tenure component of the local fertility effect. The next subsection characterizes this margin. A full general-equilibrium policy experiment also changes the price, active sets, old housing choices, fiscal transfers, and possibly the resource cost  $\Xi^\Pi$ .

## 9.2 Tenure switching at the boundary

In this subsection  $q^R$  and  $q^O$  are tenure-mode effective user costs. The model now generates its own wedge: a buyer pays  $q^O = q + \bar{\ell}/(1 + r)$  per unit while a renter pays  $q^R = q$ ; with no gains tax the modes price space identically. The anticipated tax also adds a constant to the owner case's continuation—a locked-in old household occupies space at  $q - \bar{\ell}$ , an old-age subsidy worth  $\beta\gamma \log(q/(q - \bar{\ell}))$  in value terms—which shifts tenure choice without moving any bundle. The proposition below carries all three case differences—effective price, cap, and continuation constant—explicitly; tenure-specific amenities, moving costs, or taste shifters still require a separate calculation.

Tenure switching is not a price or general-equilibrium effect. It is already present in the fixed-price model because tenure is chosen by

$$W_i^Y = \max\{W_i^R, W_i^O\}.$$

A policy that raises the owner value relative to the renter value moves households near the boundary  $W_i^O = W_i^R$  from renting to owning, and because tenure is a discrete choice, the switchers' bundles—and possibly their fertility—jump. Whether they jump depends on a single feature of the environment: whether the two tenures price a unit of space differently. With  $\bar{\ell} = 0$  they price it identically and the jump is exactly zero: the local policy formula of the previous subsection is then complete. With the tax, owning is the expensive way to occupy space, and the jump is signed accordingly; further wedges—a rental premium for large units as in the smooth-cap formulation, landlord operating costs—move the gap the same way or against it.

The case problems include savings. Concentrating the rule as above turns each case into the static problem with consumption weight  $1 + k$  and case price  $q^m$ , plus a case constant: with common old preferences and interior old choices, the owner case's continuation exceeds the renter's by

$$\Delta = \beta\gamma \log \frac{q}{q - \bar{\ell}} > 0$$

—the old-age subsidy of occupying retained space at  $q - \bar{\ell}$ —and the constants coincide when  $\bar{\ell} = 0$ . The constant moves tenure choice without moving any bundle, so the proposition carries it explicitly: write  $K^A$  and  $K^B$  for the case constants and  $\Delta = K^B - K^A \geq 0$ .

Fix prices and suppress the type index. Let  $A$  denote the cheaper case and  $B$  the more expensive case, with tenure-mode user costs

$$q^A < q^B.$$

Let  $(c_m, h_m, n_m)$  denote the case- $m$  optimum,  $m \in \{A, B\}$ . Let  $\zeta^m$  be the goods-equivalent value of relaxing the effective family-housing cap in case  $m$ .

**Proposition 3** (Fertility at a tenure switch). *Suppose  $q^A < q^B$ , the case continuations differ by  $\Delta = K^B - K^A \geq 0$ , and the household is indifferent,  $W^A = W^B$ , so the dear case runs a static-value deficit of exactly  $\Delta$ . Then:*

(i) *if the price gap dominates the constant,*

$$\alpha \log \frac{q^B}{q^A} + \vartheta \log \frac{\chi + \kappa q^B}{\chi + \kappa q^A} > \Delta,$$

*the cheaper case's effective cap binds with positive value,  $h_A = H^A$  and  $\zeta^A > 0$ ;*

(ii) *if moreover the wedge dominates the constant in resources,*

$$(q^B - q^A) h_B > \Delta (c_A - \chi n_A),$$

*the switcher upsizes and trades consumption for space:  $h_B > h_A$  and  $c_B - \chi n_B < c_A - \chi n_A$ ;*

(iii) *if in addition  $\chi \leq (1+k)\kappa q^A/\alpha$ , then  $n_B > n_A$  and  $h_B - \kappa n_B > h_A - \kappa n_A$  whenever the within-set force exceeds the level shift, the inequality displayed in the proof; with  $\Delta = 0$  the conditions in (i) and (ii) hold automatically, the level shift vanishes, and all conclusions reduce to the earlier ones.*

*Proof.* Write  $W^m = \tilde{W}^m + K^m$  up to common terms, where  $\tilde{W}^m$  is the concentrated static value at price  $q^m$  and cap  $H^m$ ; indifference is  $\tilde{W}^A - \tilde{W}^B = \Delta$ .

(i) If  $\zeta^A = 0$ , case  $A$  attains the unconstrained static value, and the closed forms give

$$\tilde{W}_{\text{unc}}(q^A) - \tilde{W}_{\text{unc}}(q^B) = \alpha \log \frac{q^B}{q^A} + \vartheta \log \frac{\chi + \kappa q^B}{\chi + \kappa q^A}.$$

Since  $\tilde{W}^B \leq \tilde{W}_{\text{unc}}(q^B)$ , the deficit would satisfy  $\Delta \geq \tilde{W}_{\text{unc}}(q^A) - \tilde{W}_{\text{unc}}(q^B)$ , contradicting price dominance. Thus  $h_A = H^A$  and  $\zeta^A > 0$ .

(ii) Suppose  $h_B \leq h_A$ . The case- $B$  bundle is then feasible in case  $A$  and costs  $(q^B - q^A)h_B$  less. The static value is concave in resources with marginal utility  $1/(c_A - \chi n_A)$  at  $w$ , so

$$\Delta = \tilde{W}^A - \tilde{W}^B \geq \frac{(q^B - q^A) h_B}{c_A - \chi n_A},$$

contradicting wedge dominance. Thus  $h_B > h_A$ .

Every case optimum satisfies

$$\frac{\vartheta}{n} = \frac{\chi}{c - \chi n} + \frac{\alpha \kappa}{h - \kappa n}.$$

Define

$$X = \frac{\chi}{c - \chi n}, \quad Y = \frac{\alpha \kappa}{h - \kappa n}.$$

$X$  is large when adult consumption is squeezed,  $Y$  when adult space is squeezed.<sup>8</sup> Then  $n = \vartheta/(X + Y)$ , and case utility can be written as  $K^m - \Phi(X, Y)$  up to a common type constant, where

$$\Phi(X, Y) = (1+k) \log X + \alpha \log Y + \vartheta \log(X + Y).$$

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<sup>8</sup>Capital  $X$  and  $Y$  are local to this proof and unrelated to income  $y_i$ .



Indifference places the two optima on level sets that differ by the constant:

$$\Phi(X_B, Y_B) = \Phi(X_A, Y_A) + \Delta.$$

Occupied housing pins down the position along a level set: at any case optimum,

$$h = \kappa \left( \frac{\alpha}{Y} + \frac{\vartheta}{X + Y} \right) = \kappa \Phi_Y(X, Y).$$

Let  $T = X + Y$ . Along a level set of  $\Phi$ ,

$$\left. \frac{d\Phi_Y}{dX} \right|_{\Phi} = \frac{\Phi_{XY}\Phi_Y - \Phi_{YY}\Phi_X}{\Phi_Y},$$

and

$$\Phi_{XY}\Phi_Y - \Phi_{YY}\Phi_X = \frac{(1+k)\alpha}{XY^2} + \frac{(1+k)\vartheta}{XT^2} + \frac{\alpha\vartheta X}{Y^2T^2} > 0,$$

so on every level set  $\Phi_Y$  is strictly increasing in  $X$ ; and at fixed  $X$ , moving to a higher level set raises  $Y$  and, since  $\Phi_{YY} < 0$ , lowers  $\Phi_Y$ . Hence  $h_B > h_A$  forces  $X_B > X_A$ , and  $c_B - \chi n_B = \chi/X_B < \chi/X_A = c_A - \chi n_A$ .

(iii) Write  $T(X; \varphi)$  for  $X + Y$  along the level set  $\Phi = \varphi$ , and decompose along the path that runs from  $(X_A, Y_A)$  along  $\Phi_A$  to  $X_B$ , then up to  $\Phi_A + \Delta$  at fixed  $X_B$ :

$$T_B - T_A = \underbrace{\int_{\Phi_A}^{\Phi_A + \Delta} \frac{d\varphi}{\Phi_Y(X_B, \cdot)}}_{\text{level shift } \geq 0} - \underbrace{\int_{X_A}^{X_B} \left. \frac{\Phi_X - \Phi_Y}{\Phi_Y} \right|_{\Phi_A} dX}_{\text{within-set force}}.$$

The within-set integrand is positive exactly where  $(1+k)Y > \alpha X$ , the region above the ray  $Y = \alpha X/(1+k)$ , which crosses each level set once: the level set is a decreasing graph, while along the ray  $\Phi = (1+k+\alpha+\vartheta)\log X + \text{constant}$  is strictly increasing in  $X$ . At a case optimum,

$$\frac{\alpha(c_m - \chi n_m)}{h_m - \kappa n_m} = q^m + \zeta^m, \quad \zeta^m \geq 0, \quad \text{so} \quad \frac{Y_m}{X_m} = \frac{\kappa(q^m + \zeta^m)}{\chi},$$

and under  $\chi \leq (1+k)\kappa q^A/\alpha$  both optima lie strictly in the region: case  $A$  because  $\zeta^A > 0$ , case  $B$  because  $q^B > q^A$ . Then  $n_B = \vartheta/T_B > \vartheta/T_A = n_A$  if and only if the within-set force exceeds the level shift. When it does,  $T_B < T_A$  together with  $X_B > X_A$  gives  $Y_B < Y_A$ , hence  $h_B - \kappa n_B = \alpha\kappa/Y_B > h_A - \kappa n_A$ . With  $\Delta = 0$  the level shift vanishes, the path stays in the region between the two optima, and the inequality always holds.  $\square$

At indifference there are now two reasons to pay more per unit of space: to get more space, and to buy the dear case's better old age. The conditions price that second temptation: the wedge must dominate the subsidy in resources for the switcher to upsize, and the fertility jump nets the within-case space force against the subsidy's level shift. At the example's parameters the conditions hold with room to spare—price dominance by a factor of four—and the boundary household upsizes from 0.15 to 0.20 and adds 0.007 children.

The rent-own cases now read off directly, with the effective user costs  $q^R = q$  and  $q^O = q + \bar{\ell}/(1+r)$ . *No gains tax* ( $\bar{\ell} = 0$ , so  $q^O = q^R = q$ ): the modes differ only through their effective caps, and an indifferent household has the same allocation in both—either both caps are slack, or the caps

coincide at the binding allocation. Then  $n_O = n_R$ , there is no discrete jump at the boundary, and the fixed-tenure formula of the previous subsection is the complete fixed-price answer. *With the tax* ( $q^O > q^R$ ): ownership is the expensive family-space mode, and an indifferent switcher into ownership upsizes and has  $n_O > n_R$ ; this is the model's own configuration for family-sized purchases. *Renting dearer* ( $q^O < q^R$ , for example a large-unit rental premium): the inequality reverses; the marginal switcher into ownership moves to a cheaper, smaller unit, and  $n_R > n_O$  under the analogous condition at the lower owner user cost. A cost wedge between the tenures is what makes tenure switching a first-order fertility-jump margin.

The switching formula is fixed-price accounting, not a welfare result or a general-equilibrium comparative static. Let types live in a smooth finite-dimensional type space with continuous density  $f(i)$ , and suppose there are no mass points at the tenure boundary. Let

$$\Delta_i(z) = W_i^O(z) - W_i^R(z), \quad \mathcal{I}(z) = \{i : \Delta_i(z) = 0\},$$

and assume  $\nabla_i \Delta_i(z) \neq 0$  on  $\mathcal{I}(z)$ , with unique case optima and kinks in active constraints having zero measure. The switching component of the fixed-price fertility effect is

$$\left. \frac{dN^{\text{switch}}}{dz} \right|_{q, \mathcal{A}} = \bar{M} \int_{\mathcal{I}(z)} \pi_i^E (n_i^O - n_i^R) \frac{\partial_z \Delta_i(z)}{\|\nabla_i \Delta_i(z)\|} f(i) dS(i),$$

where  $dS(i)$  is surface measure on the indifference boundary. For cap-shifting policies,

$$\partial_z \Delta_i(z) = \Theta_i^O \mathcal{P}_i^O(z) - \Theta_i^R \mathcal{P}_i^R(z), \quad \Theta_i^m = \lambda_i^m \zeta_i^m.$$

**Corollary 2** (Owner-access policies). *Consider an owner-access policy with  $\mathcal{P}_i^O(z) \geq 0$  and  $\mathcal{P}_i^R(z) = 0$ . Suppose the conditions of Proposition 3, including part (iii)'s inequality, hold for almost every type on  $\mathcal{I}(z)$ . If  $q^O > q^R$ , the switching component is nonnegative, so the fixed-tenure owner-access formula is a lower bound for this fixed-price fertility channel. If  $q^O = q^R$  (the baseline), the switching component is zero. If  $q^O < q^R$ , the switching component is nonpositive.*

**Corollary 3** (Rental-size limit policies). *Consider a rental-size limit policy with  $\mathcal{P}_i^R(z) \geq 0$  and  $\mathcal{P}_i^O(z) = 0$ . Suppose the conditions of Proposition 3, including part (iii)'s inequality, hold for almost every type on  $\mathcal{I}(z)$ . If  $q^O > q^R$ , the switching component is nonpositive: relaxing the rental cap raises fertility for constrained renters who remain renters, but it can also pull marginal owners back into renting. If  $q^O = q^R$  (the baseline), the switching component is zero. If  $q^O < q^R$ , the switching component is nonnegative.*

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